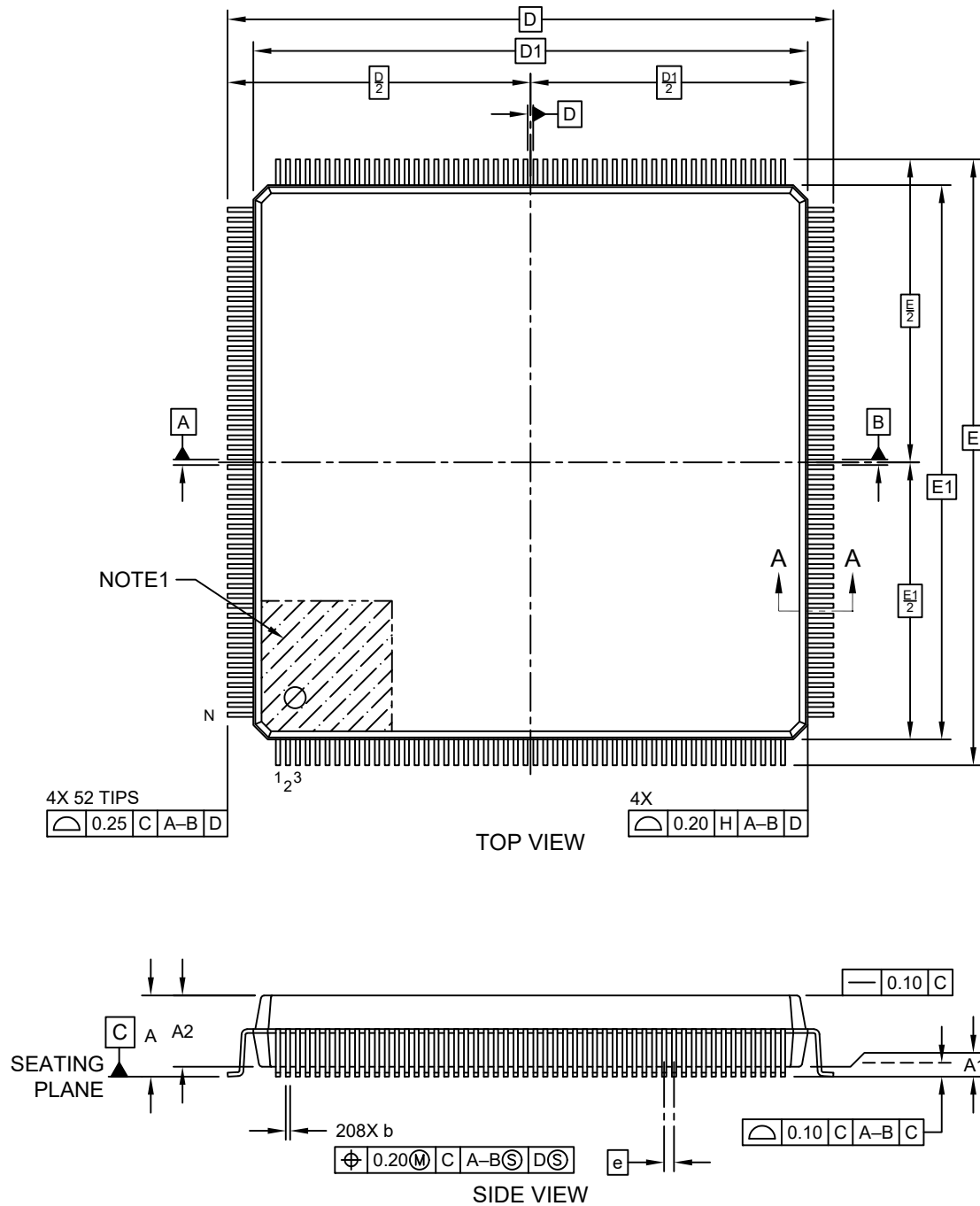


208-Lead Plastic Quad Flat Pack (KMB) - 28x28x3.4 mm Body [PQFP]

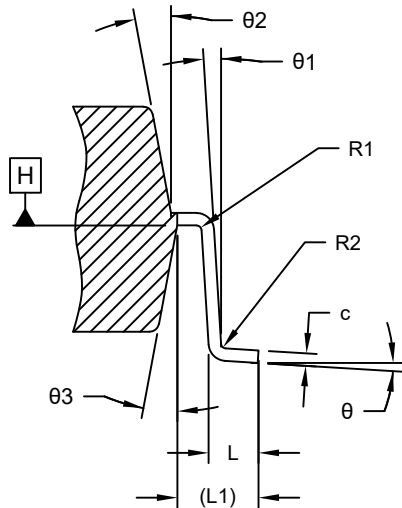
Atmel Legacy Package QAB

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>

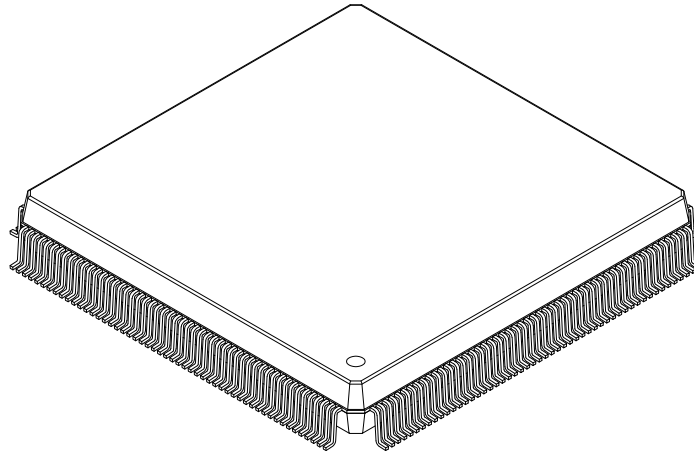


208-Lead Plastic Quad Flat Pack (KMB) - 28x28x3.4 mm Body [PQFP] Atmel Legacy Package QAB

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



SECTION A-A



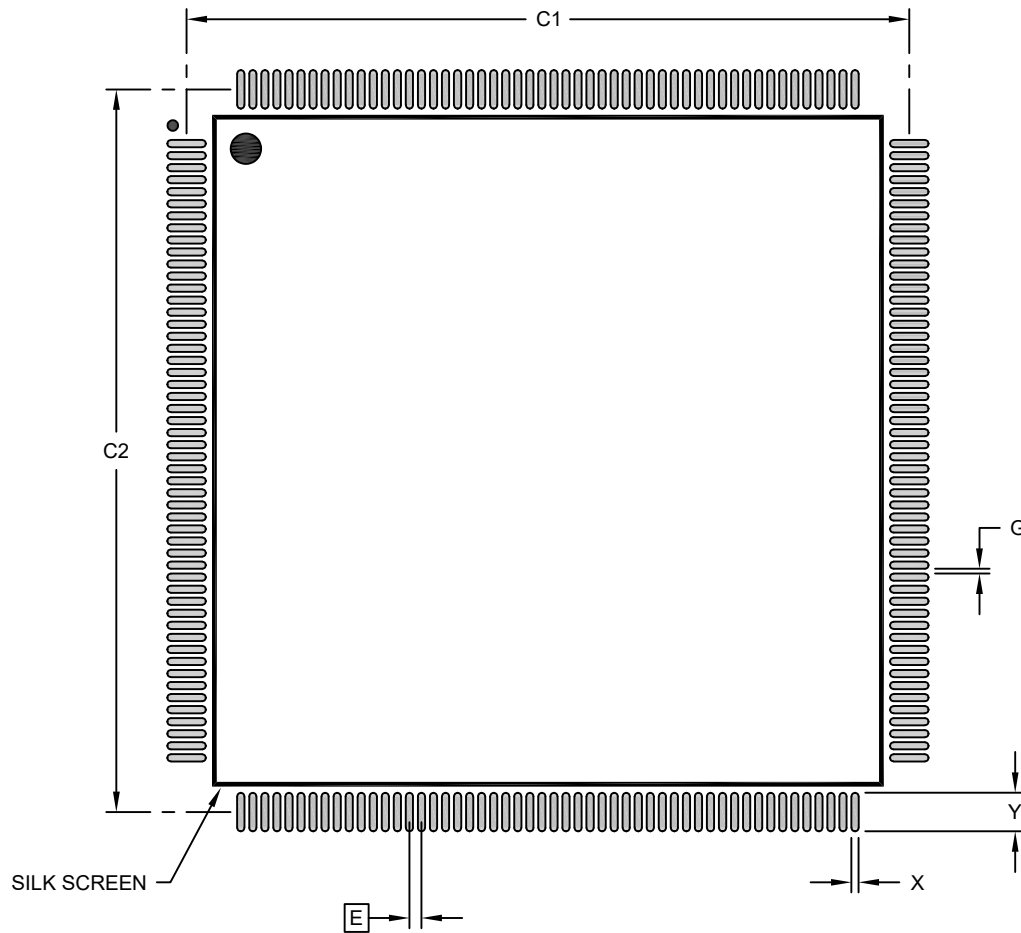
Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Number of Terminals	N	208		
Pitch	e	0.50 BSC		
Overall Height	A	—	—	4.10
Standoff	A1	0.25	—	0.50
Molded Package Thickness	A2	3.20	3.40	3.60
Overall Length	D	30.60 BSC		
Molded Package Length	D1	28.00 BSC		
Overall Width	E	30.60 BSC		
Molded Package Width	E1	28.00 BSC		
Terminal Width	b	0.17	—	0.27
Terminal Thickness	c	0.09	—	0.20
Terminal Length	L	0.45	0.60	0.75
Footprint	L1	1.30 REF		
Lead Bend Radius	R1	0.08	—	—
Lead Bend Radius	R2	0.08	—	0.25
Foot Angle	θ	0°	3.5°	8°
Lead Angle	θ1	0°	—	—
Mold Draft Angle	θ2	5°	—	16°
Mold Draft Angle	θ3	5°	—	16°

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

208-Lead Plastic Quad Flat Pack (KMB) - 28x28x3.4 mm Body [PQFP] Atmel Legacy Package QAB

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Contact Pad Spacing	C1		30.00	
Contact Pad Spacing	C2		30.00	
Contact Pad Width (Xnn)	X			0.30
Contact Pad Length (Xnn)	Y			1.60
Contact Pad to Contact Pad (Xnn)	G	0.20		

Notes:

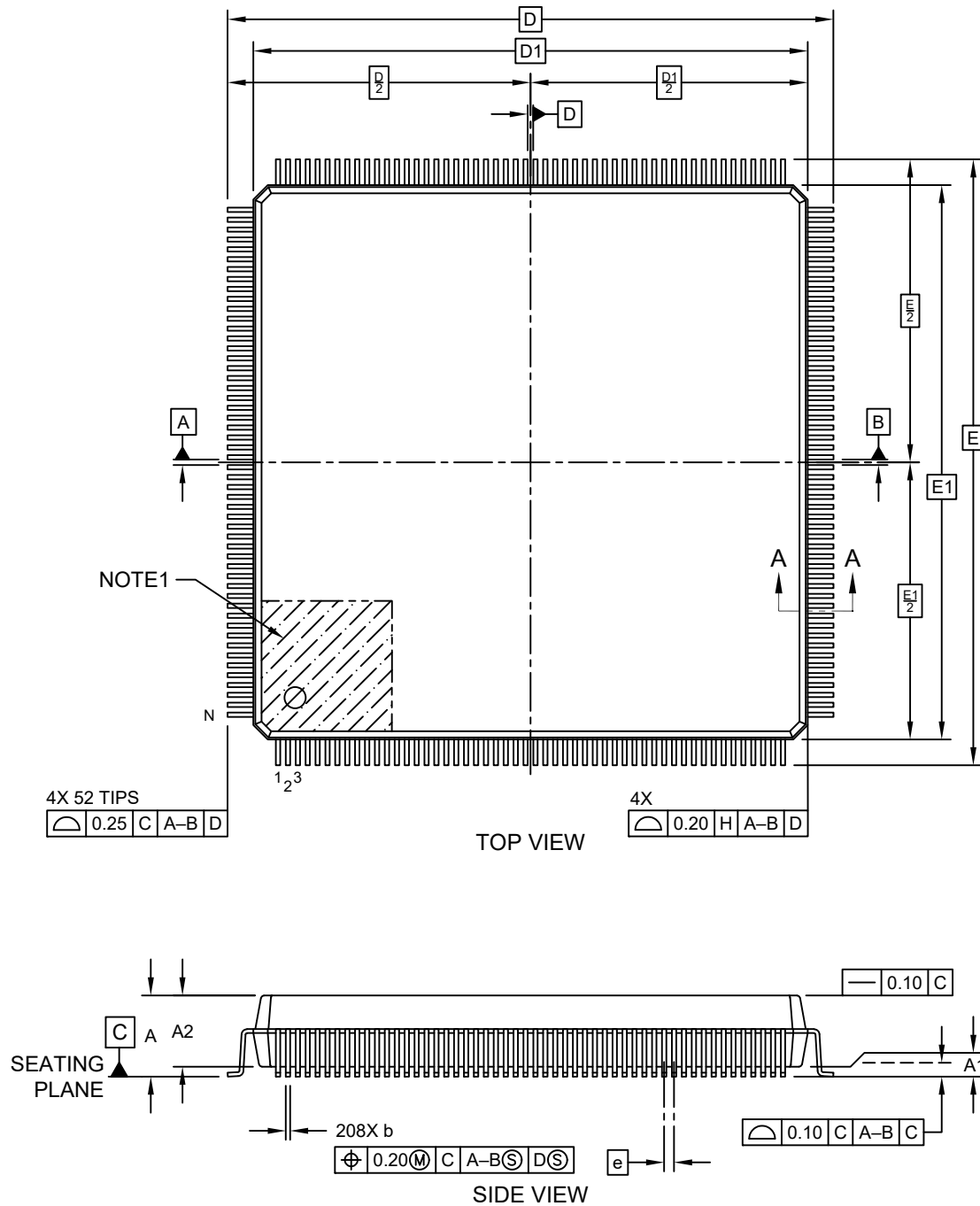
1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23294-KMB Rev A

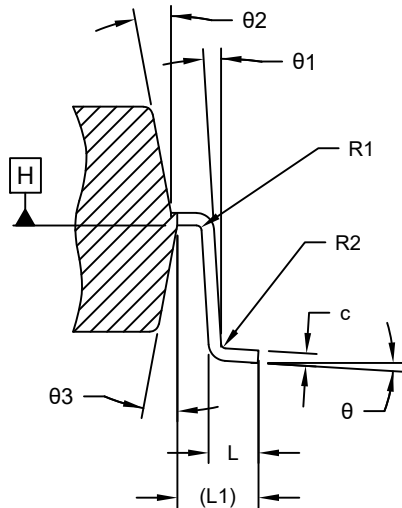
208-Lead Plastic Quad Flat Pack (KNB) - 28x28x3.4 mm Body [PQFP]
Atmel Legacy Package QAB

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/package3>

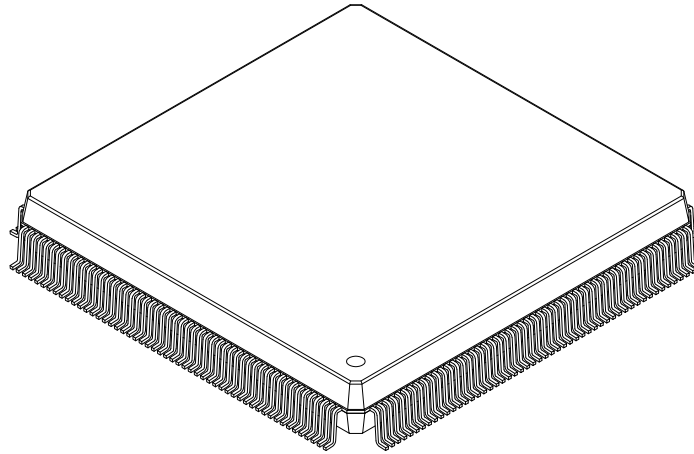


208-Lead Plastic Quad Flat Pack (KNB) - 28x28x3.4 mm Body [PQFP] Atmel Legacy Package QAB

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



SECTION A-A



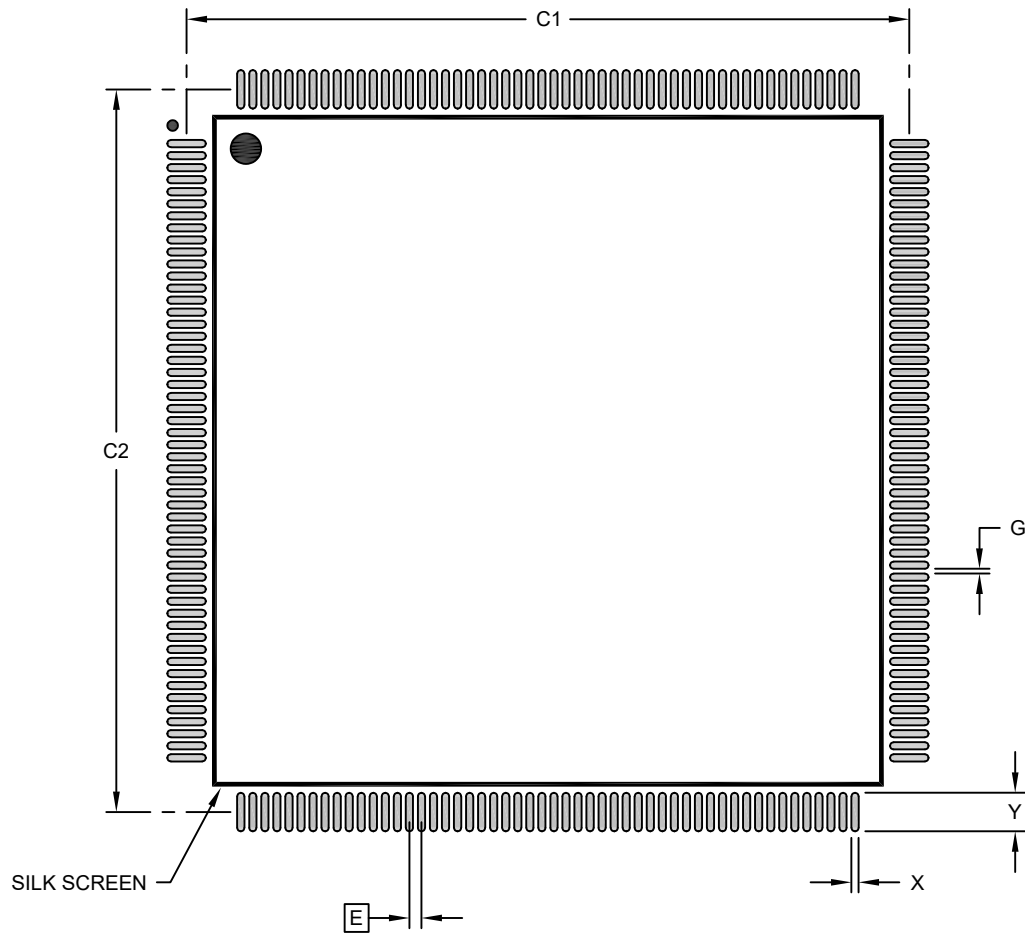
		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Terminals	N		208		
Pitch	e		0.50 BSC		
Overall Height	A		—	—	4.10
Standoff	A1		0.25	—	0.50
Molded Package Thickness	A2		3.20	3.40	3.60
Overall Length	D		30.60 BSC		
Molded Package Length	D1		28.00 BSC		
Overall Width	E		30.60 BSC		
Molded Package Width	E1		28.00 BSC		
Terminal Width	b		0.17	—	0.27
Terminal Thickness	c		0.09	—	0.20
Terminal Length	L		0.45	0.60	0.75
Footprint	L1		1.30 REF		
Lead Bend Radius	R1		0.08	—	—
Lead Bend Radius	R2		0.08	—	0.25
Foot Angle	θ		0°	3.5°	8°
Lead Angle	θ1		0°	—	—
Mold Draft Angle	θ2		5°	—	16°
Mold Draft Angle	θ3		5°	—	16°

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

208-Lead Plastic Quad Flat Pack (KNB) - 28x28x3.4 mm Body [PQFP] Atmel Legacy Package QAB

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Contact Pad Spacing	C1		30.00	
Contact Pad Spacing	C2		30.00	
Contact Pad Width (Xnn)	X			0.30
Contact Pad Length (Xnn)	Y			1.60
Contact Pad to Contact Pad (Xnn)	G	0.20		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23294-KNB Rev A